Question 1

+4 -1

A charge of $4\ \mu C$ is to be divided into two parts. The distance between the two divided charge is constant. The magnitude of the divided charges so that the force between them is maximum, will be:

(A) $1\mu C$ and $3\mu C$

(B) $2\mu C$ and $2\mu C$

(C) 0 and $4\mu C$

(D) $1.5\mu C$ and $2.5\mu C$

- A
- B
- C
- D

Question Palette

1 2 3 4 5
6 7 8 9 10
11 12 13 14 15
16 17 18 19 20
21 22 23 24

Question 2

+4 -1

Force between two identical spheres charged with same charge is F. If 50% charge of one sphere is transferred to the other sphere then the new force will be

(A) $\frac{3}{4}\ \mathrm{F}$

(B) $\frac{3}{8}F$

(C) $\frac{3}{2}F$

(D) none of these

- ◯ A
- ◯ B
- ◯ C
- ◯ D

Question Palette

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	

Question 3

+4 -1

Force between two point charges (q_1) and (q_2) placed in vacuum at 'r' cm apart is F. Force between them when placed in a medium having dielectric constant $K = 5$ at '$r/5$' cm apart will be:

(A) $F/5$
(B) 25 F
(C) 5 F
(D) $F/25$

- A
- B
- C
- D

Question Palette

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	

Question 4

+4 -1

Find the number of electron that should be removed from a coin so that it acquires a charge of 10^{-6} coulomb

(A) 6.25×10^{11}

(B) 1.6×10^{6}

(C) 1.6×10^{-19}

(D) 6.25×10^{12}

- A
- B
- C
- D

Question Palette

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	

Question 5

+4 -1

Quantisation of charge implies:

(A) charge cannot be destroyed
(B) charge exists on particles
(C) there is a minimum permissible charge on a particle
(D) charge, which is a fraction of a coulomb is not possible.

- A
- B
- C
- D

Question Palette

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	

Question 6

+4 -1

Electric charge of any system is:

(A) zero or neutral

(B) a half integral multiple of the minimum amount of charge

(C) an integral multiple of the minimum amount of charge

(D) Twice the least amount of charge

- A
- B
- C
- D

Question Palette

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	

Question 7

+4 -1

The minimum possible charge on an object is

(A) 1 coulomb
(B) 1 stat coulomb
(C) 1.6×10^{-19} coulomb
(D) 3.2×10^{-19} coulomb

- A
- B
- C
- D

Question Palette

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	

Question 8

+4 -1

One million electrons are added to a glass rod. The total charge on the rod is:

(A) $10^{-13}\,\mathrm{C}$

(B) $-1.6 \times 10^{-13}\,\mathrm{C}$

(C) $+1.6 \times 10^{-12}\,\mathrm{C}$

(D) $10^{-12}\,\mathrm{C}$

- ◯ A
- ◯ B
- ◯ C
- ◯ D

Question Palette

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	

Question 9

+4 -1

The sum of mole fractions of binary solution is always:

(A) greater than 1
(B) equal to 1
(C) less than 1
(D) equal to 0

- A
- B
- C
- D

Question Palette

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	

Question 10

+4 -1

The density (in g mL^{-1}) of a 3.6 M sulphuric acid solution, i.e., 29% H_2SO_4 (molar mass = 98 g mol^{-1}) by mass will be;

(A) 3.45
(B) 2.64
(C) 1.88
(D) 1.22

- A
- B
- C
- D

Question Palette

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	

Question 11

+4 -1

The molarity of NaOH solution if 250 mL of it contains 1 mg of NaOH is;

(A) 10^{-1} M

(B) 10^{-2} M

(C) 10^{-4} M

(D) 10^{-3} M

- A
- B
- C
- D

Question Palette

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	

Question 12

+4 -1

Which of the following is dependent on temperature?

(A) Molarity
(B) Mole fraction
(C) Mass percentage
(D) Molality

- A
- B
- C
- D

Question Palette

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	

Question 13

+4 -1

A solution is prepared by adding 5 g of a substance (X) to 18 g of water. Calculate the mass percentage of the solute:

(A) 21.74%
(B) 19%
(C) 25%
(D) 40%

- A
- B
- C
- D

Question Palette

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	

Question 14

+4 -1

Which of the following is not a binary solution?

(A) Pure water + Sugar
(B) Air
(C) Mixture of benzene and toluene
(D) Mixture of ethanol and methanol

- A
- B
- C
- D

Question Palette

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	

Question 15

+4 -1

Calculate the volume by volume percentage in the solution of 15 ml of alcohol in 75 ml of water;

(A) 1.66
(B) 166.6
(C) 16.67
(D) 17.99

- A
- B
- C
- D

Question Palette

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	

Question 16

+4 -1

Match **List-I** with **List-II** and choose the **correct** option.

List-I		List-II	
A.	Soda water	I.	Solution of gas in solid
B.	Sugar solution	II.	Solution of gas in gas
C.	Air	III.	Solution of gas in liquid
D.	Hydrogen gas in palladium	IV.	Solution of solid in liquid

(A) A-III, B-IV, C-II, D-I
(B) A-IV, B-III, C-II, D-I
(C) A-III, B-IV, C-I, D-II
(D) A-IV, B-I, C-III, D-II

- A
- B

Question Palette

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	

Question 17 +4 -1

The determinant

$$\begin{vmatrix} -a+b+c & -2a & -2a \\ -2b & -b+c+a & -2b \\ -2c & -2c & -c+a+b \end{vmatrix}$$ is

(1) 0

(2) a perfect square

(3) a perfect cube

(4) none of these

- A
- B

Question Palette

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	

Question 18

+4 -1

If α, β and γ are the roots of the equation $x^2(px+q) = r(x+1)$. Then the value of determinant $\begin{vmatrix} 1+\alpha & 1 & 1 \\ 1 & 1+\beta & 1 \\ 1 & 1 & 1+\gamma \end{vmatrix}$ is

(A) $\alpha\beta\gamma$

(B) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$

(C) 0

(D) $\frac{1}{\alpha\beta\gamma}$

- A
- B
- C

Question Palette

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	

Question 19

+4 -1

If $D = \begin{vmatrix} 1 & 2 & 3 \\ 2 & -1 & 0 \\ 3 & 4 & 5 \end{vmatrix}$, then $\begin{vmatrix} 1 & 6 & 3 \\ 4 & -6 & 0 \\ 3 & 12 & 5 \end{vmatrix}$ is equal to

(A) $2D$
(B) $3D$
(C) $6D$
(D) 0

- A
- B
- C

Question Palette

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	

Question 20

+4 -1

Solution set of the equation $\begin{vmatrix} x & 3 & 7 \\ 2 & x & 2 \\ 7 & 6 & x \end{vmatrix} = 0$ is:

(A) $\{-9, 2, 7\}$
(B) {0,2,7}
(C) {1,2,7}
(D) none of these

- A
- B
- C
- D

Question Palette

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	

Question 21

+4 -1

If $\begin{vmatrix} k & -2 \\ 5 & 7 \end{vmatrix} = \begin{vmatrix} 1 & -1 \\ 2 & 3 \end{vmatrix}$, then k is:

(A) $\frac{-5}{7}$

(B) $\frac{5}{7}$

(C) $\frac{-3}{2}$

(D) $\frac{3}{2}$

- A
- B
- C
- D

Question Palette

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	

Question 22

+4 -1

Let $ax^3 + bx^2 + cx + d = \begin{vmatrix} 3x & x+1 & x-1 \\ x-3 & -2x & x+2 \\ x+3 & x-4 & 5x \end{vmatrix}$

,

then the value of d is

(A) 5

(B) 0

(C) -6

(D) 4

- A
- B
- C

Question Palette

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	

Question 23

+4 -1

$$\begin{vmatrix} a+b & a+2b & a+3b \\ a+2b & a+3b & a+4b \\ a+4b & a+5b & a+6b \end{vmatrix} =$$

(A) $a^2+b^2+c^2-3abc$

(B) $3ab$

(C) $3a+5b$

(D) 0

- ◯ A
- ◯ B
- ◯ C
- ◯ D

Question Palette

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	

Question 24

+4 -1

The roots of the equation $\begin{vmatrix} x-1 & 1 & 1 \\ 1 & x-1 & 1 \\ 1 & 1 & x-1 \end{vmatrix} = 0$, are

(A) 1,2

(B) $-1, 2$

(C) $1, -2$

(D) $-1, -2$

- A
- B
- C
- D

Question Palette

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	